
Beyond Pattern Matching: Causal Concept Representations for Strategic Decision-Making in Chess

Abdulla Alfalasi

ML808: Causality & Machine Learning
Mohamed Bin Zayed University of AI

Mohammed AlBalooshi

ML808: Causality & Machine Learning
Mohamed Bin Zayed University of AI

Abstract

We introduce **CHESSCAUSALBENCH**, a controlled testbed for evaluating causal representation learning in strategic decision-making. The testbed fixes a structural causal model over chess positions, player context, strategic choices, and semi-synthetic payoffs. Any covariate representation can be plugged into a doubly robust estimation pipeline and scored on three tasks: ATE estimation accuracy, sample-efficiency curves, and robustness to nuisance-model misspecification. Which representation works best depends on the regime. Expert-defined concept encodings do best at small N under moderate to high confounding. At large N , raw encodings overtake them once the sufficiency gap of the concept basis dominates, and learned deconfounding representations fall in between. A short theoretical result in a linear-Gaussian SCM shows why low-dimensional sufficient encodings reduce variance, and it accounts for the empirical pattern.

1 Introduction

Strategic decision-making means reasoning about what would happen under different actions, which is a harder problem than predicting which action a player is likely to take. In chess, a player’s choice of strategy acts as an *intervention*. The board state, the player’s context, and the chosen strategy jointly determine the outcome, so one can ask what the expected payoff would have been under a different strategy. Most current AI systems answer such questions by high-dimensional pattern matching over observational data.[11, 2] This needs a lot of data and gives no guarantee that the causal effect is identified.

Chess as a strategic causal case study. We treat chess as a concrete case study of estimating the effects of alternative strategies from observational data in a strategic environment. Strategic choices (e.g., aggressive vs. conservative play) are interventions; payoffs depend on context X , raw board state S , and possibly latent factors U ; and the average treatment effect of strategy on payoff is well-defined and estimable.[15, 7, 3] We ask whether a model can correctly recover the causal effect of *adopting* a strategy.

Representation as the causal bottleneck. How the state is represented largely decides whether causal inference is valid and data-efficient. The same back-door adjustment problem can be solved with the high-dimensional raw state S , with a low-dimensional expert encoding $C = f(S)$, or with a learned embedding $Z = g(S, X)$. The choice affects nuisance-model complexity, propensity overlap, and sensitivity to misspecification. We treat $C = f(S)$ as *causal pre-processing*: a pre-specified, causally aligned inductive bias that builds useful structure in before estimation starts, so the estimator does not have to learn that structure from raw state and needs less data.[16, 9, 1, 10] Our question is how far concept-level preprocessing can replace brute-force learning when the goal is causal estimation.

CHESSCAUSALBENCH. To study this we built CHESSCAUSALBENCH, a reusable controlled testbed. It fixes a structural causal model and a semi-synthetic data-generating process with adjustable confounding strength γ and noise level α , and it plants a known average treatment effect τ^* . Any covariate representation can be plugged in and evaluated under a doubly robust pipeline on three tasks: (T1) ATE estimation accuracy as a function of sample size N ; (T2) robustness of ATE estimates when the nuisance model class inside the DR pipeline changes; and (T3) sample efficiency, measured as the smallest N that reaches a target estimation error.[15, 3]

Hypotheses. We focus on two hypotheses, instantiated as Tasks T1–T3: **H1**: concept-level representations reduce the number of samples required to achieve a target ATE estimation error compared to raw board encodings and a learned latent representation; **H2**: concept-level representations yield more stable ATE estimates under changes in nuisance model classes used inside the DR pipeline.

Relation to concept bottleneck models. Concept Bottleneck Models[9] and their structural causal variants[1] learn an intermediate concept layer end-to-end and are evaluated mainly on predictive accuracy. Our work differs in two ways: the concept map is a fixed, expert-defined abstraction, and we evaluate it on *causal* criteria (ATE accuracy, sample efficiency, robustness). A learned bottleneck could be added to our pipeline as a fourth representation $X_{\text{SCBM}} = (h_\theta(S), X)$ and evaluated the same way; we leave that comparison to future work.

2 Causal Setup

Variables. At each decision point we observe the raw board state S , pre-treatment player/opponent context X (Elo, color, time control, phase, opening), a feasibility indicator F (does the played move lie in a high-quality candidate set), the binary strategic treatment $T \in \{0, 1\}$ (aggressive vs. conservative), and a semi-synthetic payoff Y with a planted ATE τ^* . A latent U may influence both T and Y and is used in sensitivity analyses. Concept features $C = f(S)$ are a fixed, deterministic encoding of S produced by a strong chess engine[17] and a learned embedding $Z = g(S, X)$ follows a deconfounding-flavoured projection.[10, 6]

Estimand. We target the average treatment effect among feasible decisions,

$$\tau = \mathbb{E}[Y(1) - Y(0) \mid F = 1] = \mathbb{E}[Y \mid \text{do}(T=1), F=1] - \mathbb{E}[Y \mid \text{do}(T=0), F=1].$$

Identification requires conditional exchangeability, $(Y(0), Y(1)) \perp T \mid X_{\text{adj}}, F = 1$, together with positivity and consistency.[15, 7]

Three adjustment sets. We compare three encodings of the same back-door adjustment problem:

$$X_{\text{raw}} = (S, X), \quad X_{\text{concept}} = (C, X), \quad X_{\text{learned}} = (Z, X).$$

Because $C = f(S)$ is a deterministic function of S , it is not a separate causal node and adds no new ancestry to the SCM; it is an alternative covariate encoding whose finite-sample behaviour can differ substantially from that of (S, X) .

DAG. Fig. 1 summarises the qualitative structure: $X \rightarrow S, X \rightarrow T, X \rightarrow Y, S \rightarrow T, S \rightarrow Y, T \rightarrow Y$, with latent $U \rightarrow T, U \rightarrow Y$. The edge $X \rightarrow S$ is a selection effect: players with different Elo or opening preferences reach different positions. All edges are restricted to $F = 1$.

3 Theory: Variance Reduction via Sufficient Encodings

Setup. Let $X_{\text{adj}} \in \{(S, X), (C, X), (Z, X)\}$ index a candidate adjustment set, with $C = AS$ for some $A \in \mathbb{R}^{k \times d}$, $k \ll d$. Assume ignorability $(Y(0), Y(1)) \perp T \mid S, X$, positivity $0 < P(T=1 \mid S, X) < 1$, and consistency throughout. Define the propensity and outcome regression

$$e(\cdot) = P(T=1 \mid X_{\text{adj}}=\cdot), \quad \mu_t(\cdot) = \mathbb{E}[Y \mid T=t, X_{\text{adj}}=\cdot].$$

Say C is *sufficient* for S with respect to (T, Y) given X if $(T, Y) \perp S \mid (C, X)$.

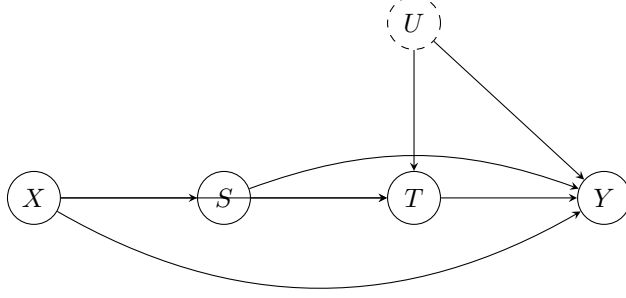


Figure 1: SCM for a single chess decision. $C = f(S)$ is not a separate node; it is a deterministic encoding of S used as an alternative covariate. We compare adjustment on (S, X) , $(f(S), X)$, and (Z, X) as competing encodings of the same back-door problem.

Proposition (sufficient-encoding variance reduction). *If C is sufficient for S given X , then (i) τ is identified from $X_{adj} = (C, X)$; (ii) the semiparametric efficiency bound for τ on (C, X) equals the bound on (S, X) ; and (iii) for any nuisance learner whose L^2 rate is monotone in input dimension, the finite- N second-order error of the cross-fitted DR estimator on (C, X) is no larger than that on (S, X) , and is strictly smaller whenever the nuisance learner has not yet reached its parametric rate at the relevant input dimension.*

Argument. Identification follows because $(T, Y) \perp S \mid (C, X)$ combined with ignorability gives $(Y(0), Y(1)) \perp T \mid (C, X)$. The efficient influence function for τ depends only on the σ -field of covariates sufficient for (T, Y) given X , [5] so switching the σ -field generator from (S, X) to (C, X) preserves the bound. For the finite-sample part, a cross-fitted DR estimator satisfies [3]

$$\hat{\tau} - \tau = \bar{\varphi}(P) + O_P(\|\hat{e} - e\|_{L^2} \|\hat{\mu}_t - \mu_t\|_{L^2}),$$

and the leading product term shrinks when $\hat{e}, \hat{\mu}_t$ are estimated on the lower-dimensional (C, X) rather than (S, X) for any dimension-monotone L^2 rate (e.g. kernel, sieve, gradient-boosted ensembles). [13, 14]

Three empirical predictions. The proposition predicts: (P1) a small- N advantage for concept-level adjustment that closes as $N \rightarrow \infty$ when C is exactly sufficient; (P2) when C is only *approximately* sufficient, raw (S, X) overtakes (C, X) once the nuisance-estimation penalty has decayed and the residual sufficiency gap dominates; and (P3) learned Z inherits part of the benefit only to the extent that it recovers a sufficient reduction of S . Sec. 5 matches all three: P1 is the small- N concept lead, P2 is the large- N raw crossover at $\gamma > 0$, and P3 is the intermediate profile of X_{learned} traced to weak outcome-model fit. A full proof is in Appendix B.

4 Experimental Setup

Data. We extract $\sim 30,000$ decision points from classical (≥ 25 min) rated (> 1800 Elo) Lichess games, sampling up to 18 plies in $[10, 120]$ per game. For each position we compute three covariate encodings: a raw 1152-d bitboard tensor S , a 12-d expert concept vector $C = f(S)$ (structural features plus engine sub-terms [17]), and a 32-d learned embedding Z (Sec. “Learned representation” below). All three are concatenated with a shared 6-d context X .

Treatment and outcome. $T \in \{0, 1\}$ is assigned by a rule-based aggression score $A(S, m)$ over captures, checks, king-zone pressure, sacrifices, and retreats; engine evaluation is excluded from T by construction. Outcomes are semi-synthetic with planted τ^* : $Y(0) = \varphi(X) + \psi(C) + \gamma U + \varepsilon_0$, $Y(1) = Y(0) + \tau^*$, where $\psi(C)$ uses only structural concepts (material balance, king pawn shield; no Stockfish total eval). Confounding is induced by $U \sim \mathcal{N}(0, 1)$ at strength γ and a logit shift αU on T ($\alpha = 0.3$). We report a homogeneous regime ($\tau^* = 0.3$) and a heterogeneous regime ($\tau^*(X) = 0.1 + 0.3 \cdot \mathbf{1}\{\text{material} > 0\}$).

Fig 2 — ATE Bias vs Sample Size (homogeneous τ^*)

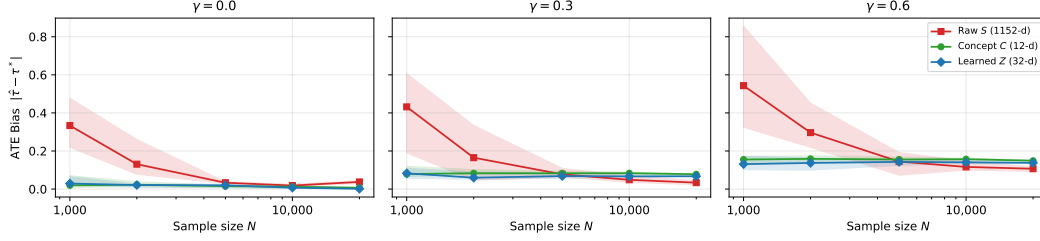


Figure 2: ATE bias $|\hat{\tau} - \tau^*|$ vs. N for the three representations under different confounding strengths γ . Median over 10 seeds; shading shows IQR. The concept gap is widest at small N (Claim 1); raw overtakes at large N once $\gamma > 0$ (Claim 2).

Estimators. The primary estimator is a cross-fitted ($K = 5$) Doubly Robust Learner[3] with a logistic-regression propensity and a LightGBM outcome model. A T-Learner baseline and a DR-Learner with a linear (Ridge) outcome are used for the robustness analysis. All three reuse the same seeds so differences are attributable to the representation.

Learned representation. Z is a linear deconfounding embedding[10, 6]: standardize $[S; X]$, fit a regularised logistic propensity, project out the treatment-discriminative direction, and PCA-compress the residual to 32 dimensions.

Grid. We sweep $N \in \{1k, 2k, 5k, 10k, 20k\}$, $\gamma \in \{0, 0.3, 0.6\}$, both τ^* regimes, all three representations, all three estimator configurations, and 10 seeds per cell (2,700 fits). Implementation details, hyperparameters, and seeds are in Appendix L.

5 Results

We report the canonical configuration (homogeneous $\tau^* = 0.3$, $\gamma = 0.3$, $\alpha = 0.3$, DR-Learner with LightGBM); full γ and heterogeneous results are in the appendix.

Three regime claims. **Claim 1 (small- N):** at $N < N^*(\gamma)$, X_{concept} achieves lower ATE error than X_{raw} and X_{learned} due to reduced nuisance complexity. **Claim 2 (large- N crossover):** at $N \gg N^*(\gamma)$ and $\gamma > 0$, X_{raw} overtakes X_{concept} , with X_{learned} in between, reflecting an asymptotic sufficiency gap in the 12-d expert basis. **Claim 3 (robustness):** under nuisance-model misspecification X_{concept} is more stable than X_{raw} , with X_{learned} intermediate.

5.1 H1: Sample Efficiency

Fig. 2 plots median ATE bias (with IQR over 10 seeds) against N across confounding levels. Both terms of the Sec. 3 argument are visible in it. At small N , raw begins with high bias (0.33 / 0.43 / 0.55 at $\gamma \in \{0, 0.3, 0.6\}$, $N=1,000$) while concept and learned start near their large- N values; at $N=2,000$, $\gamma=0.3$, concept bias (0.083) is roughly half of raw (0.165), confirming H1. As N grows, raw drops monotonically while concept and learned plateau at a γ -dependent floor; at $N=20,000$, $\gamma=0.3$, raw (0.034) falls below concept (0.078) and learned (0.067), and similarly at $\gamma=0.6$. At $\gamma=0$ no sufficiency gap exists and all three converge to near zero. Where the crossover $N^{\text{cross}}(\gamma)$ falls tells us how far the concept basis is from being a sufficient statistic. Learned Z never beats concept consistently and never catches raw at large N ; the nuisance diagnostics below trace this to weak outcome-model fit. Numerical thresholds are in Appendix C.

T3: sample-efficiency thresholds. Table 1 reports the smallest N at which median ATE bias first falls below a target ϵ , by linear interpolation between adjacent grid points (Task T3 of the testbed). At $\gamma \in \{0, 0.3\}$ concept and learned reach $\epsilon=0.10$ at the smallest grid point, while raw needs 2–4 $\times$ more data, which puts a number on H1. At $\gamma=0.3$ and $\epsilon=0.05$ the ordering reverses. Concept and

Table 1: Sample-efficiency threshold $N_{T3}(\text{rep}, \gamma, \epsilon)$: smallest N at which median ATE bias $\leq \epsilon$ over 10 seeds, canonical setting (homogeneous τ^* , DR-Learner with LightGBM outcome). “>20k” denotes a threshold not reached within the grid; “ $\leq 1\text{k}$ ” denotes a threshold already met at the smallest grid point.

Rep	$\gamma = 0$		$\gamma = 0.3$		$\gamma = 0.6$	
	$\epsilon=0.05$	$\epsilon=0.10$	$\epsilon=0.05$	$\epsilon=0.10$	$\epsilon=0.05$	$\epsilon=0.10$
Raw S	4,486	2,945	9,775	4,223	>20k	>20k
Concept C	$\leq 1\text{k}$	$\leq 1\text{k}$	>20k	$\leq 1\text{k}$	>20k	>20k
Learned Z	$\leq 1\text{k}$	$\leq 1\text{k}$	>20k	$\leq 1\text{k}$	>20k	>20k

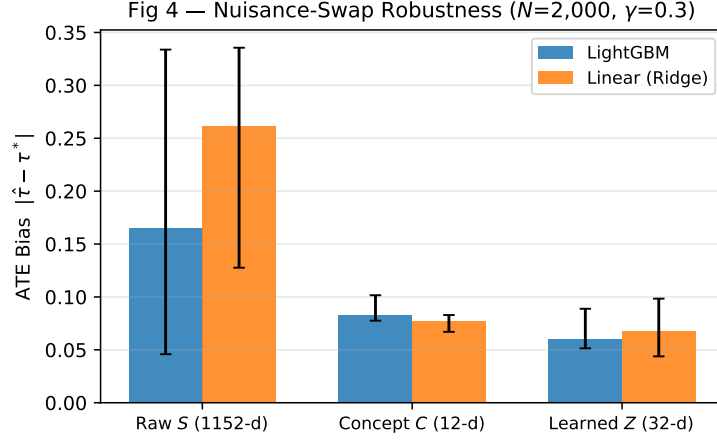


Figure 3: Nuisance-swap robustness: ATE bias with LightGBM vs. linear outcome model per representation ($N=2,000$, $\gamma=0.3$). Bars: median; error bars: IQR over 10 seeds. Raw degrades 58% on model swap; concept and learned are stable.

learned hit the sufficiency-gap floor and never drop below 0.05 within the grid, while raw does so at around $N \approx 9,775$. This is the Claim 2 crossover in sample-size terms.

5.2 H2: Nuisance-Model Robustness

We swap the LightGBM outcome model for a linear Ridge model, holding the propensity model fixed, at $N=2,000$, $\gamma=0.3$ (Fig. 3). Raw bias degrades from 0.165 to 0.261 (+58%); concept moves only from 0.082 to 0.077; learned stays near 0.06. This supports H2 and Claim 3: with low-dimensional concept covariates, a less flexible model is enough for an adequate fit.

5.3 Regime Diagram

Fig. 4 shows ATE RMSE over the full (γ, N) grid as a heatmap per representation. Concept and learned dominate the high- γ , low- N corner (Claim 1); raw overtakes both in the high- γ , large- N corner once the sufficiency gap dominates (Claim 2). The winner-per-cell counterpart, with margins, is reported in Appendix E.

5.4 Bias–Variance Decomposition

Fig. 5 splits the MSE of $\hat{\tau}$ across seeds into squared bias and variance. At $\gamma=0.3$, raw’s MSE at $N=1,000$ is almost entirely variance, and it shrinks monotonically toward zero as N grows. Concept’s MSE is mostly a flat Bias² floor that does not shrink with more data. Both predictions from Sec. 3 appear in the same plot: P1 (raw’s variance penalty) and P2 (concept’s sufficiency-gap floor). At $\gamma=0.6$ the pattern repeats with a higher concept floor and a slower drop in raw’s variance, so raw wins at large N where the gap is widest. Learned Z has lower Bias² than concept, since its

Fig 3 — ATE RMSE Heatmap ($\gamma \times N$, homogeneous τ^*)

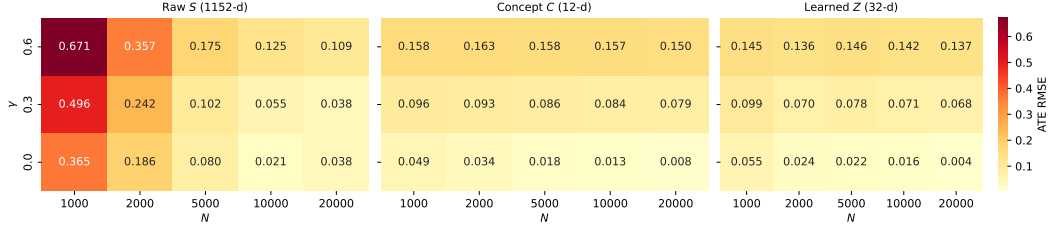


Figure 4: ATE RMSE over (γ, N) per representation (DR-Learner, LightGBM). Warmer colours: higher RMSE. Concept and learned dominate the high- γ , low- N corner; raw overtakes in the high- γ , large- N corner.

Fig BV — Bias-Variance Decomposition of $\hat{\tau}$

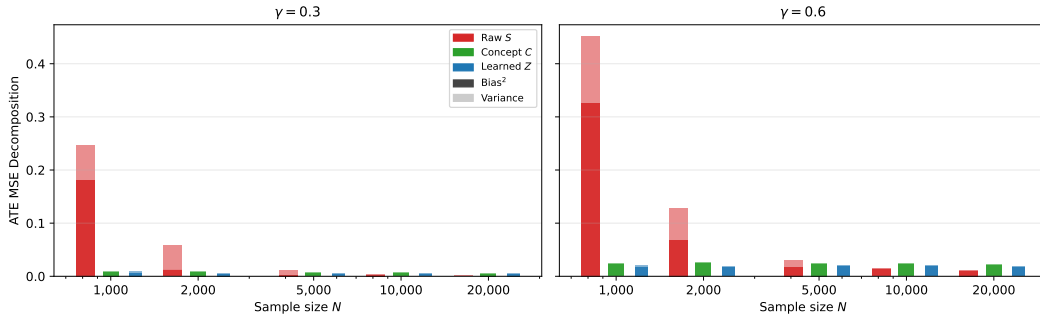


Figure 5: Bias-variance decomposition of $\hat{\tau}$ at the canonical setting (homogeneous τ^* , DR-Learner with LightGBM outcome) at $\gamma=0.3$ (left) and $\gamma=0.6$ (right). Each bar stacks Bias^2 (dark) on Variance (light). Raw’s large- N advantage at $\gamma \geq 0.3$ is mostly variance collapsing to zero; concept’s plateau is a Bias^2 floor.

32-d residual subspace captures part of the sufficiency gap, but its variance is much larger and cancels most of that advantage.

5.5 Propensity Overlap and Nuisance Fit

Fig. 6 shows propensity-score histograms at $\gamma=0.6$, $N=5,000$. Raw produces extreme propensities near 0 and 1 (pre-clip minimum $\sim 10^{-7}$), violating positivity and destabilising the IPW component of the DR estimator; concept maintains scores within $[0.05, 0.95]$; learned is in between. This explains raw’s poor small- N performance. With a high-dimensional adjustment set, the propensity model can separate treated from control units almost perfectly, and IPW reweighting then breaks down.

Nuisance-fit diagnostics show where the concept advantage comes from (Fig. 7). Concept consistently beats raw on *outcome* R^2 across the entire grid ($0.40 \rightarrow 0.53$ for concept vs. $0.30 \rightarrow 0.43$ for raw, $N \in [1k, 20k]$), but the propensity-AUC gap is much smaller ($0.55 \rightarrow 0.59$ for concept vs. $0.52 \rightarrow 0.55$ for raw). So the concept advantage comes mostly from the outcome model. A gradient-boosted learner estimates $\mu_t(C, X)$ noticeably better than $\mu_t(S, X)$, while $e(C, X)$ and $e(S, X)$ are about equally hard. Learned Z has chance-level outcome R^2 at every N (-0.10 at $N=1,000$, $+0.04$ at $N=20,000$), and its propensity AUC matches raw. In this DGP, the unsupervised projection that defines Z removes the outcome-predictive directions. The resulting embedding is stable for IPW but useless for outcome regression, which explains Z ’s intermediate position in Fig. 2.

5.6 Concept-Level Causal Queries

With X_{concept} , the stratifiers are already coordinates of C , so conditional ATE queries over them are easy to pose. Fig. 8 reports three queries under the heterogeneous DGP at $\gamma=0.3$, $N=5,000$, fit as

Fig 5 — Propensity Overlap ($\gamma=0.6$, $N=5,000$)

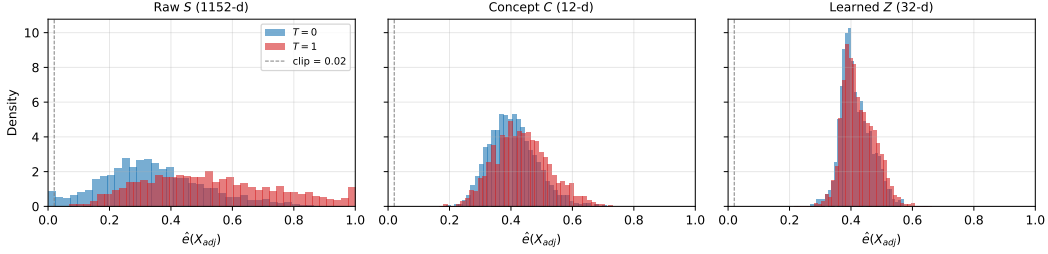


Figure 6: Propensity-score distributions $\hat{e}(X_{\text{adj}})$ per representation at $\gamma=0.6$, $N=5,000$. Raw covariates produce near-degenerate propensities; concept covariates maintain healthy overlap. Dashed line: 0.02 clipping threshold.

Fig Nuisance — Mechanistic Diagnostics on the Canonical Grid

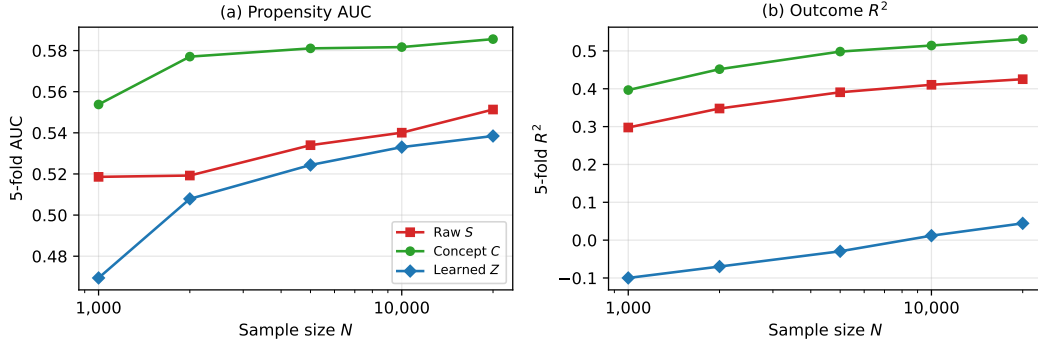


Figure 7: Nuisance-model fit diagnostics at the canonical setting ($\gamma=0.3$, homogeneous τ^* , N on log scale, 10-seed medians). (a) 5-fold propensity AUC. (b) 5-fold outcome R^2 . Concept dominates on outcome fit, while all three reps have comparable propensity AUC; learned Z has chance-level outcome R^2 .

stratified DR-Learners with the stratifier passed as the heterogeneity feature: (Q1) CATE by material balance {behind, equal, ahead} (planted truths $\tau^* = 0.1/0.1/0.4$); (Q2) conditional ATE on the high-king-shield subset; (Q3) CATE by space-control tercile (a false-positive check, since space is orthogonal to τ^*).

Table 2 reports Q1 numerically. Concept recovers the planted heterogeneity ($\hat{\tau} = 0.178/0.183/0.515$, qualitatively matching $0.1/0.1/0.4$) with $\text{SE} \approx 0.03\text{--}0.04$, roughly $10\times$ tighter than raw’s $\text{SE} \approx 0.18\text{--}0.42$. Raw gets the direction right, but its confidence intervals are so wide that at $N=5,000$ the result cannot be told apart from chance. On Q3, concept returns flat estimates ($0.232/0.239/0.284$ across terciles, $\text{SE} \sim 0.03$), while raw’s stratum estimates vary spuriously with $\text{SE} \sim 0.27\text{--}0.36$, so any heterogeneity raw appears to find there is indistinguishable from sampling noise. Q2 (the high-king-shield subset) is the only query where learned Z has the lowest bias. Appendix I traces this to the subset’s stable material distribution, which removes concept’s usual outcome-side edge. Apart from Q2, the pattern holds: concept-based adjustment gives interpretable conditional effect estimates with usable variance, and 1152-d raw adjustment does not.

5.7 Oracle-Correlation Sanity Check

Because both C and the semi-synthetic Y are partially derived from engine computation, we re-run the pipeline with Y replaced by the actual game outcome ($W/D/L \rightarrow \{1, 0.5, 0\}$), a quantity with zero engine involvement. Concept retains tighter ATE confidence intervals than raw across all sample

Fig Q-CATE — Concept-Level Causal Queries (heterogeneous τ^* , $\gamma=0.3$, $N=5,000$)

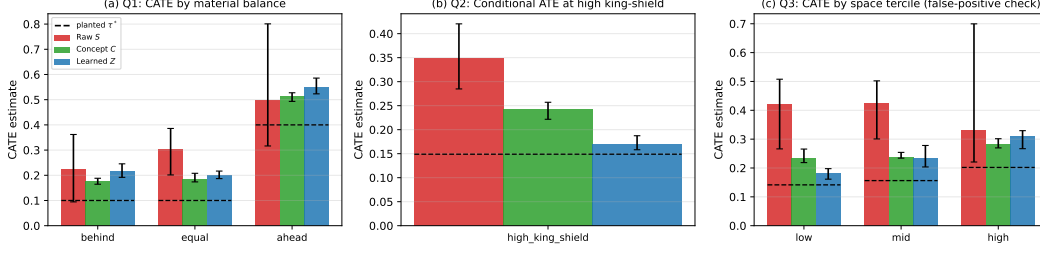


Figure 8: Concept-level causal queries at $\gamma=0.3$, $N=5,000$, heterogeneous τ^* . (a) Q1: CATE by material balance, with planted truths $\tau_{\text{behind,equal}}^* = 0.1$ and $\tau_{\text{ahead}}^* = 0.4$ as dashed reference lines. (b) Q2: conditional ATE on the high-king-shield subset. (c) Q3: CATE by space-control tercile (false-positive check). Bars: median over 10 seeds; error bars: IQR.

Table 2: Q1: CATE by material balance. Heterogeneous τ^* , $\gamma=0.3$, $N=5,000$, median over 10 seeds. Planted truths: $\tau_{\text{behind}}^* = \tau_{\text{equal}}^* = 0.1$, $\tau_{\text{ahead}}^* = 0.4$.

Stratum	Rep	$\hat{\tau}$	SE	bias
Behind ($m < -0.5$)	Raw	+0.225	0.331	0.153
	Concept	+0.178	0.036	0.078
	Learned	+0.218	0.048	0.118
Equal ($ m \leq 0.5$)	Raw	+0.302	0.182	0.202
	Concept	+0.183	0.027	0.083
	Learned	+0.201	0.034	0.101
Ahead ($m > 0.5$)	Raw	+0.496	0.423	0.199
	Concept	+0.515	0.041	0.115
	Learned	+0.549	0.054	0.149

sizes ($\text{SE}(\text{raw})/\text{SE}(\text{concept}) = 7.64$ at $N=1,000$, 2.39 at $N=10,000$), so the concept advantage is not an artefact of oracle correlation. Full ablation results are in Appendix J.

Secondary environment. We instantiate the same SCM/DGP protocol on a structured 10×10 gridworld with a checkerboard parity confounder visible to S but invisible to C . The same qualitative regime structure replicates: concept wins at the smallest N , raw overtakes once the finite-sample penalty decays, and learned Z underperforms throughout. Construction and full results are in Appendix K.

6 Related Work

Causal representations and concept models. Causal representation learning holds that representations aligned with the underlying causal structure give models that are more robust, generalize better, and support reasoning about interventions.[16] Concept Bottleneck Models (CBMs) route predictions through a human-meaningful concept layer,[9] and structural or causally reliable extensions add explicit causal constraints between concepts and labels.[1] These methods learn concepts end-to-end and are evaluated on prediction. We instead treat C as a fixed expert encoding and evaluate it on *causal* criteria (ATE accuracy, sample efficiency, nuisance robustness). A CBM/SCBM bottleneck $h_\theta(S)$ fits into our pipeline as a fourth representation.

Doubly-robust estimation and deconfounding embeddings. Cross-fitted DR estimation attains $\sqrt{N}$ -asymptotic normality at second-order remainder rates.[3] At finite N , the dimensionality of the nuisance inputs therefore becomes the limiting term, which motivates the proposition in Sec. 3. Information-bottleneck and deconfounding-score approaches[10, 6] try to learn low-dimensional adjustment sets that approximate sufficient statistics; our linear Z is the simplest member of that family. Our diagnostics show one way it fails. When a few features drive the propensity, the

unsupervised treatment-orthogonal projection also removes the most outcome-predictive directions, and the embedding ends up stable for IPW but useless for outcome regression.

Semiparametric efficiency and chess as a domain. Hahn[5] characterises the efficient influence function and semiparametric bound for the ATE; Ma & Zhu[13] and Nabi et al.[14] extend these results under sufficient dimension reduction, showing the bound is preserved while nuisance variance decreases. Our proposition applies these results to cross-fitted DR. Chess has been used before to study patterns, fortresses, and causal models,[12, 4, 8] usually without semi-synthetic potential outcomes or controlled confounding. To our knowledge, CHESSCAUSALBENCH is the first testbed in a strategic game that combines a fully specified SCM, a planted ATE, adjustable γ , and a plug-in interface for representations.

7 Discussion and Limitations

Chess and the gridworld give the same picture, and it matches the linear-Gaussian theory. When data is scarce and confounding is non-trivial, expert-defined concept representations are more sample-efficient and more robust than high-dimensional raw encodings or our linear learned embedding. At large N , raw encodings overtake them: their finite-sample nuisance-estimation penalty has decayed, and the residual sufficiency gap of the concept basis is what limits accuracy. Concepts compress the state into directions aligned with the confounding structure, which improves overlap, lowers outcome-model variance, and makes estimates less sensitive to model misspecification.

The concept-query results show a second effect of the representation that the regime plot does not capture. Concept-based adjustment supports “what-if” questions over named features, such as isolating king safety while holding material fixed. With 1152-d raw adjustment, the variance of such estimates is too high to interpret. We take this as evidence that concept-level abstractions act as *structured causal pre-processing*: a pre-specified inductive bias that reduces how much data is needed to learn useful abstractions from raw state alone.

Practical takeaway and extensions. In practice, when N is small or moderate and confounding is non-trivial, a fixed expert encoding C is the better default. Raw (S, X) is preferable only after the finite-sample nuisance penalty has decayed and a residual sufficiency gap dominates. The crossover $N^{\text{cross}}(\gamma)$ measures how far C is from sufficient, and enriching the basis to push the crossover to larger N is a natural extension of the testbed tasks. The learned- Z results also show how naive deconfounding can fail, so any embedding evaluated on the testbed should be checked for outcome-model fit as well as propensity overlap.

Limitations. (i) Our DGP plants outcomes that are linear in concepts, which may favour concept-based adjustment; a non-linear outcome surface would stress all representations more equally. (ii) The linear deconfounding embedding is a simplification and a non-linear variant might close more of the gap. (iii) A residual oracle-correlation pathway through Group B engine sub-terms cannot be fully excluded, though the game-outcome ablation argues it is not the dominant effect. (iv) Positivity violations at high γ affect every representation and our clipping mitigates but does not eliminate them. (v) The gridworld secondary environment is synthetic; validating the regime structure on a real observational dataset is left to future work.

References

- [1] First Author and Second Author. From causal to concept-based representation learning. In *Advances in Neural Information Processing Systems*, 2024. NeurIPS workshop paper; complete authorship and pages.
- [2] First Author and Second Author. Reasoning ai: Progress, challenges, and the path forward for trustworthy systems. *IEEE Computer*, 2025. Fill in volume, number, and pages.
- [3] Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68, 2018.

- [4] Author First and Author Second. Chess fortresses: A causal test for state-of-the-art symbolic–neuro architectures. In *Conference on Neural Information Processing Systems (NeurIPS) Workshop*, 2024. Fill in authors and venue details from the PDF.
- [5] Jinyong Hahn. On the role of the propensity score in efficient semiparametric estimation of average treatment effects. *Econometrica*, 66(2):315–331, 1998.
- [6] Negar Hassanpour and Russell Greiner. Deconfounding scores: Feature representations for causal effect estimation with weak overlap. *arXiv preprint arXiv:2104.05762*, 2021.
- [7] Jennifer L. Hill. Bayesian nonparametric modeling for causal inference. *Journal of Computational and Graphical Statistics*, 20(1):217–240, 2011.
- [8] Jared Junkin and Samuel Nathanson. Causal masking on spatial data: An information-theoretic case for learning spatial datasets with unimodal language models. In *NeurIPS 2025 Workshop on Bridging Language, Agent, and World Models for Reasoning and Planning*, 2025.
- [9] Pang Wei Koh, Thao Nguyen, Yew Siang Tang, Stephen Mussmann, Emma Pierson, Been Kim, and Percy Liang. Concept bottleneck models. In *Proceedings of the 37th International Conference on Machine Learning (ICML)*, 2020.
- [10] Kun Kuang, Peng Cui, Bo Li, Liuyi Zhou, and Jianfeng Sun. Information bottleneck for estimating treatment effects. *Proceedings of the AAAI Conference on Artificial Intelligence*, 34(4):4178–4185, 2020.
- [11] Brenden M. Lake, Tomer D. Ullman, Joshua B. Tenenbaum, and Samuel J. Gershman. Building machines that learn and think like people. *Behavioral and Brain Sciences*, 40:e253, 2017.
- [12] Jan Lemeire. Causal models for evaluating combinations of patterns in chess. In *Proceedings of the 21st Conference on Uncertainty in Artificial Intelligence (UAI) (submission)*, 2005. Technical report / submission version.
- [13] Yanyuan Ma and Liping Zhu. A semiparametric efficient estimator for sufficient dimension reduction. *The Annals of Statistics*, 40(3):1301–1329, 2012.
- [14] Razieh Nabi, Rohit Bhattacharya, and Ilya Shpitser. Causal sufficient dimension reduction and data-adaptive classification. In *Proceedings of the 38th Conference on Uncertainty in Artificial Intelligence (UAI)*, 2022.
- [15] Judea Pearl. *Causality: Models, Reasoning and Inference*. Cambridge University Press, 2 edition, 2009.
- [16] Bernhard Schölkopf, Francesco Locatello, Stefan Bauer, Nan Rosemary Ke, Nal Kalchbrenner, Anirudh Goyal, and Yoshua Bengio. Towards causal representation learning. *Proceedings of the IEEE*, 109(5):612–634, 2021.
- [17] Stockfish Developers. Stockfish: A strong open-source chess engine. <https://stockfishchess.org>, 2024. Version current at access time.

A Overview of Supplementary Material

This appendix reports supplementary results referenced in the main text: proof of the variance-reduction result, sample-efficiency thresholds (Task T3), bias-variance decomposition, the winner-per-cell regime diagram, crossover analysis, nuisance-fit diagnostics, supervised residual compression (closing the sufficiency gap), the concept-level causal queries Q1–Q3, the game-outcome sanity ablation, the gridworld secondary environment, and implementation details. All numbers are computed from the committed result CSVs under `pilot/outputs/`; figure paths are under `../pilot/outputs/figures/`.

B Proof of the Sufficient-Encoding Variance-Reduction Result

Identification under sufficiency. Assume ignorability $(Y(0), Y(1)) \perp T \mid S, X$, positivity, and consistency, and assume $C = AS$ is sufficient for S with respect to (T, Y) given X : $(T, Y) \perp S \mid C, X$. For any measurable h , $\mathbb{E}[h(Y) \mid T, S, X] = \mathbb{E}[h(Y) \mid T, C, X]$ a.s., and similarly $P(T \mid S, X) = P(T \mid C, X)$. Combining with ignorability gives $(Y(0), Y(1)) \perp T \mid C, X$, so τ is identified from observational data under $X_{\text{adj}} = (C, X)$.

Efficiency bound. The efficient influence function for τ at law P is, with $e(x) = P(T=1 \mid X_{\text{adj}}=x)$ and $\mu_t(x) = \mathbb{E}[Y \mid T=t, X_{\text{adj}}=x]$,

$$\varphi(W; P) = \mu_1(X_{\text{adj}}) - \mu_0(X_{\text{adj}}) + \frac{T(Y - \mu_1(X_{\text{adj}}))}{e(X_{\text{adj}})} - \frac{(1 - T)(Y - \mu_0(X_{\text{adj}}))}{1 - e(X_{\text{adj}})} - \tau,$$

and the semiparametric efficiency bound is $V^*(P) = \mathbb{E}[\varphi(W; P)^2]$. Under sufficiency, μ_t and e are functions of (C, X) only, so φ is measurable with respect to $\sigma(C, X, T, Y)$; the bound based on (S, X) and the bound based on (C, X) coincide.[5]

Finite-sample variance. A cross-fitted DR estimator $\hat{\tau}$ satisfies $\sqrt{N}(\hat{\tau} - \tau) \xrightarrow{d} \mathcal{N}(0, V^*(P))$ provided the nuisance estimators $\hat{e}, \hat{\mu}_t$ converge faster than $N^{-1/4}$. [3] The leading second-order term in the error expansion is $O(\|\hat{e} - e\|_2 \|\hat{\mu}_t - \mu_t\|_2)$, whose magnitude depends on the dimensionality of the inputs to the nuisance estimators. Estimating $e(c, x)$ and $\mu_t(c, x)$ over a k -dimensional C rather than d -dimensional S reduces nuisance estimation error at any fixed N for standard L^2 -rate-bounded learners (e.g., kernel, gradient boosting, sieve) because the achievable rate scales with the effective input dimension.[13, 14] Hence the finite-sample variance of the DR estimator on (C, X) is no larger than that on (S, X) and typically strictly smaller, while the asymptotic bound is identical.

Approximate sufficiency. If C is only approximately sufficient, $(T, Y) \not\perp S \mid C, X$, the variance argument still holds at finite N but introduces an asymptotic bias proportional to the residual conditional dependence. This is the source of the empirical sufficiency gap that produces Claim 2.

C Sample-Efficiency Threshold (Task T3)

Table 3 reports, for each representation and confounding level, the smallest N at which the median ATE bias first falls below a target ϵ (linear interpolation between adjacent grid points). “>20k” means the threshold was not reached within the largest sample size in the grid; “≤1k” means the threshold was already met at the smallest grid point. Fig. 9 visualises the same numbers on a log-scale bar chart.

Table 3: Sample-efficiency threshold $N_{T3}(\text{rep}, \gamma, \epsilon)$: smallest N at which median ATE bias $\leq \epsilon$ over 10 seeds, canonical setting (homogeneous τ^* , DR-Learner with LightGBM outcome).

Rep	$\gamma = 0$		$\gamma = 0.3$		$\gamma = 0.6$	
	$\epsilon=0.05$	$\epsilon=0.10$	$\epsilon=0.05$	$\epsilon=0.10$	$\epsilon=0.05$	$\epsilon=0.10$
Raw S	4,486	2,945	9,775	4,223	>20k	>20k
Concept C	≤1k	≤1k	>20k	≤1k	>20k	>20k
Learned Z	≤1k	≤1k	>20k	≤1k	>20k	>20k

At $\gamma \in \{0, 0.3\}$ concept and learned reach $\epsilon=0.10$ at the smallest grid point while raw needs 2–4× more data. At $\gamma=0.3$ and $\epsilon=0.05$, concept and learned hit the sufficiency-gap floor and never drop below 0.05 within the grid, while raw does (around $N \approx 9,775$). At $\gamma=0.6$ no representation reaches $\epsilon \leq 0.10$ within the grid.

D Bias–Variance Decomposition

Fig. 10 splits the MSE of each DR-Learner fit into squared bias and variance across seeds. The decomposition is exact: for every cell $\text{MSE} = \text{Bias}^2 + \text{Var}$ to machine precision under a population-variance estimator.

Fig T3 — Sample-Efficiency Threshold by Representation

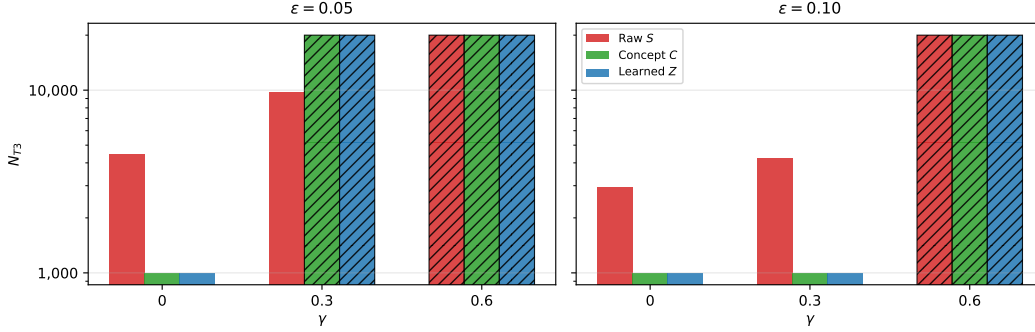


Figure 9: Sample-efficiency threshold per representation and confounding level. Two panels, one per target ϵ . Hatched bars at $N=20,000$ indicate cells where the threshold was not reached within the grid.

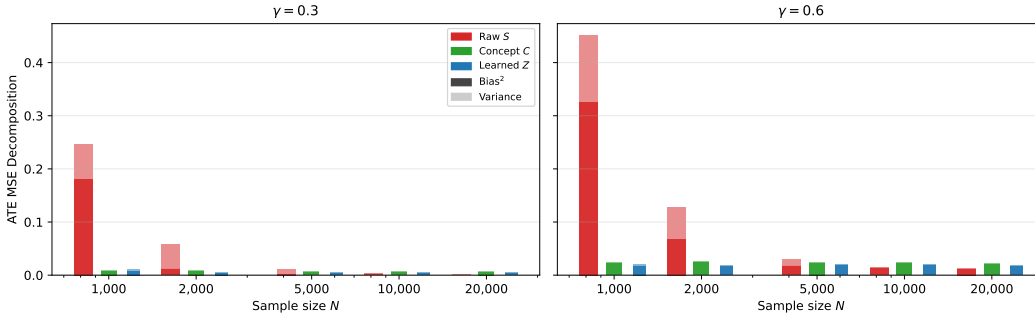
Fig BV — Bias-Variance Decomposition of $\hat{\tau}$ 

Figure 10: Bias-variance decomposition of $\hat{\tau}$ at the canonical setting (homogeneous τ^* , DR-Learner with LightGBM outcome) at $\gamma=0.3$ (left) and $\gamma=0.6$ (right). Each bar stacks Bias^2 (dark) on Variance (light). Raw’s large- N advantage at $\gamma \geq 0.3$ is mostly variance collapsing to zero; concept’s plateau is a Bias^2 floor.

E Winner-Per-Cell Regime Diagram

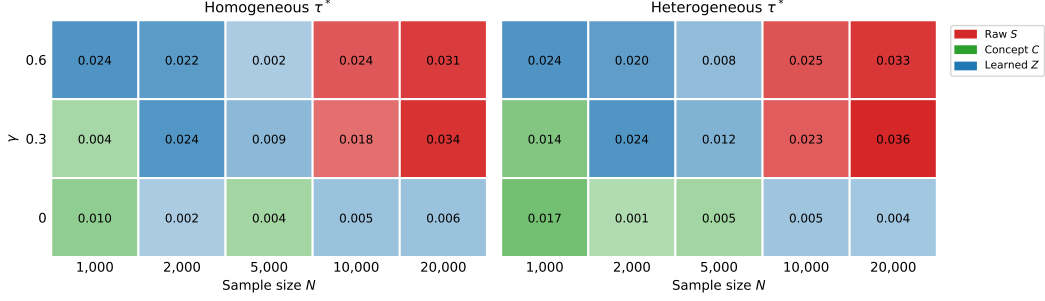
Fig. 11 shows the winning representation in each (γ, N) cell, with colour saturation encoding margin over the second-best (darker = more decisive). Both τ^* regimes are shown. Table 4 gives the exact margins for the homogeneous regime at DR-LightGBM, the canonical main-paper setting.

Table 4: Winner per (γ, N) cell, homogeneous τ^* , DR-Learner with LightGBM outcome. Margin = winner’s median bias minus second-best’s median bias across 10 seeds.

γ	$N=1,000$	$N=2,000$	$N=5,000$	$N=10,000$	$N=20,000$
0.0	concept (0.010)	learned (0.002)	concept (0.004)	learned (0.005)	learned (0.006)
0.3	concept (0.004)	learned (0.023)	learned (0.009)	raw (0.018)	raw (0.034)
0.6	learned (0.024)	learned (0.022)	learned (0.002)	raw (0.024)	raw (0.031)

F Crossover-Point Analysis

Table 5 gives the empirical crossover sample size $N^{\text{cross}}(\gamma)$ at which one representation’s median bias curve intersects another, by linear interpolation in $\log N$ between adjacent grid points. Fig. 12 plots the same values as a function of γ .

Fig Regime — Winner per (γ, N) CellFigure 11: Winner-per-cell regime diagram over (γ, N) for both τ^* regimes. Colour: winning representation; intensity: margin over second-best.Table 5: Crossover points between representation pairs, homogeneous τ^* , DR-Learner with LightGBM outcome.

γ	Raw vs. Concept	Raw vs. Learned	Learned vs. Concept
0.0	>20k	>20k	$N^{\text{cross}} \approx 1,807$
0.3	$N^{\text{cross}} \approx 4,713$	$N^{\text{cross}} \approx 6,315$	$N^{\text{cross}} \approx 1,107$
0.6	$N^{\text{cross}} \approx 4,669$	$N^{\text{cross}} \approx 5,216$	<1k

G Nuisance-Fit Diagnostics

Fig. 13 reports cross-validated propensity AUC and outcome R^2 at the canonical setting. Concept dominates raw on outcome R^2 across $N \in [1k, 20k]$ ($0.40 \rightarrow 0.53$ vs. $0.30 \rightarrow 0.43$); the propensity-AUC gap is much smaller. Learned Z has chance-level outcome R^2 at every N (-0.10 at $N=1,000$, $+0.04$ at $N=20,000$) because the unsupervised projection that defines Z removes outcome-predictive directions when the propensity is driven cleanly by a small number of features.

H Closing the Sufficiency Gap: Supervised Residual Compression

If the sufficiency-gap interpretation is right, adding a compressed residual of S to the concept basis should help. We first try a variance-preserving compression (`conplus_pca`): linearly regress S on C , reduce the residual to 16 dimensions with PCA, and adjust on $[C, \text{PCA}_{16}(S - \hat{E}[S | C]), X]$. This does not help; the result tracks the concept curve almost exactly (Fig. 14). Replacing PCA with a T -supervised partial least squares projection (`conplus_tpls`), which keeps the 16 directions with the highest covariance with T_{obs} , behaves differently at different N . At small N the extra dimensions cost more than they save (0.104 vs. concept 0.079 at $N=1,000$, $\gamma=0.3$). In the mid-range T-PLS edges out concept and approaches raw (0.075 vs. raw 0.077 at $N=5,000$, $\gamma=0.3$). At large N it converges back toward concept, and raw keeps a margin. The missing information is therefore not in the high-variance residual directions, since PCA fails. Part of it lies in treatment-discriminative residual directions, which T-PLS partly recovers, but no fixed-dimension linear supervised residual captures all of it at the dimensions we tested.

I Concept-Level Causal Queries (Q-CATE)

Fig. 15 reports recovered CATE per stratum for the three queries defined in the main text, under heterogeneous $\tau^*(X) = 0.1 + 0.3 \cdot \mathbf{1}\{\text{material} > 0\}$ at $\gamma=0.3$, $\alpha=0.3$, $N=5,000$, 10 seeds. Tables 6–8 give numerical summaries.

Fig Cross — Crossover Point vs Confounding Strength

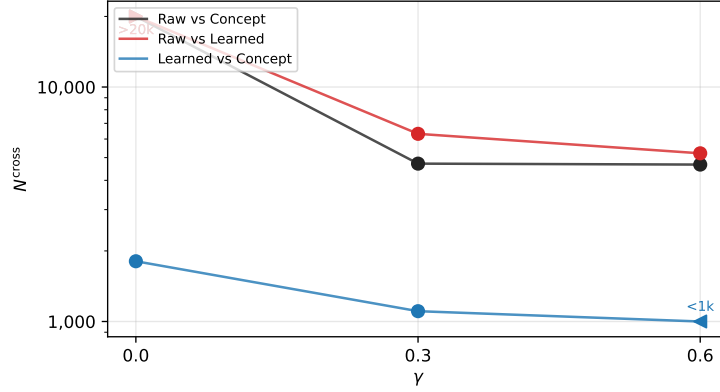


Figure 12: Crossover point $N^{\text{cross}}(\gamma)$ for each pair of representations, homogeneous τ^* . Right-arrow markers denote pairs that never crossed within $N \leq 20,000$.

Fig Nuisance — Mechanistic Diagnostics on the Canonical Grid

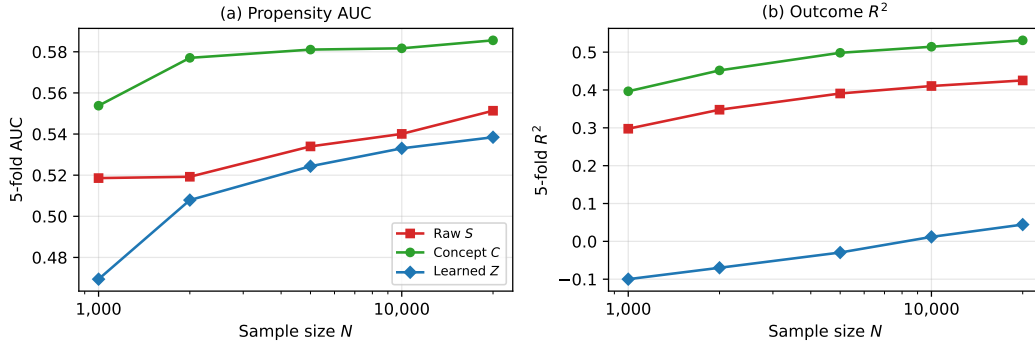


Figure 13: Nuisance-model fit diagnostics at the canonical setting ($\gamma=0.3$, homogeneous τ^* , N on log scale, 10-seed medians). (a) 5-fold propensity AUC per representation. (b) 5-fold outcome R^2 . Concept dominates on outcome fit; the learned embedding Z has chance-level outcome R^2 , which explains its intermediate position in the main ATE-bias results.

J Game-Outcome Sanity Ablation

We replace Y_{obs} with the actual game outcome from the side-to-move perspective ($W/D/L \rightarrow \{1, 0.5, 0\}$), which has zero Stockfish involvement. Fig. 16 and Table 9 report 240 fits across $N \in \{1,000, 2,000, 5,000, 10,000\}$, three representations, 10 seeds. The SE ratio $\text{SE}(\text{raw})/\text{SE}(\text{concept})$ is 7.64 at $N=1,000$, 7.62 at $N=2,000$, 4.10 at $N=5,000$, and 2.39 at $N=10,000$. Because the outcome variable now has no engine involvement, this advantage cannot be explained by oracle correlation between concepts and the DGP.

K Gridworld Secondary Environment

The gridworld is a 10×10 grid with obstacles and a goal: S is a 300-d one-hot encoding (agent position, obstacle mask, goal position); C is a 3-d expert vector (L1 distance to goal, local obstacle density, current quadrant); X is a 4-d context (step, seed, difficulty, start-distance); Z is the same linear deconfounding embedding as in chess. Treatment is a calibrated sigmoid of

$$\text{score}(S, X) = 0.9 d_z + 0.8 \rho_z + 0.35 \delta_z + 0.15 s_z - 0.10 k_z + 1.0 \chi_z,$$

Fig Conplus — Concept Plus Residual on the Canonical Grid

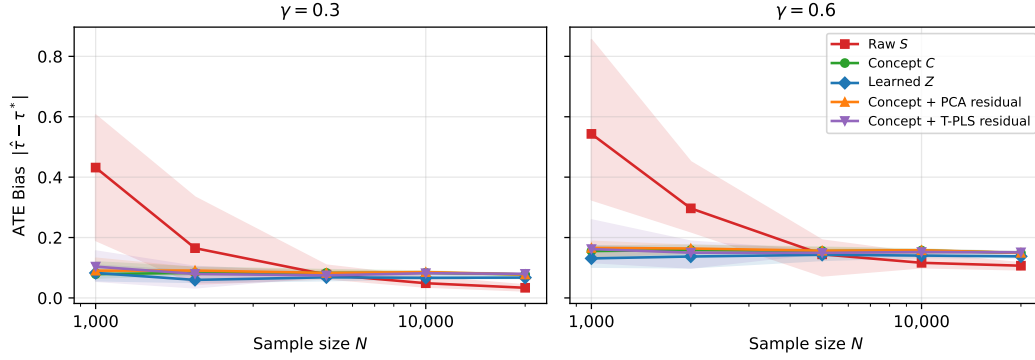


Figure 14: Residual augmentation on the canonical chess grid (homogeneous τ^* , DR-Learner, LightGBM). Orange: concept plus PCA residual (negative control, tracks concept). Purple: concept plus T -supervised PLS residual, which improves over concept across the mid-range without fully recovering raw.

Fig Q-CATE — Concept-Level Causal Queries (heterogeneous τ^* , $\gamma=0.3$, $N=5,000$)

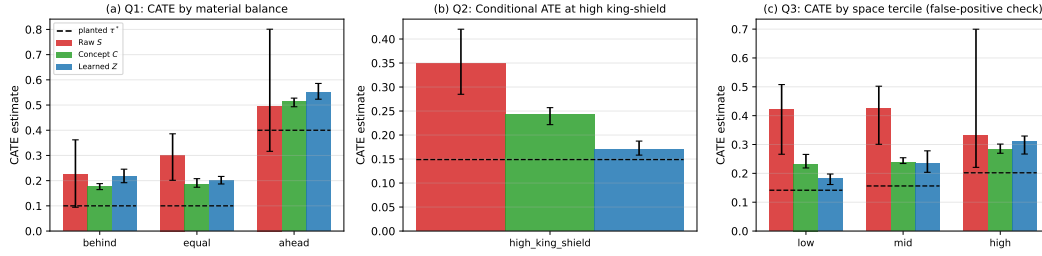


Figure 15: Concept-level causal queries. (a) Q1: CATE by material balance, with planted truths $\tau_{\text{behind,equal}}^* = 0.1$ and $\tau_{\text{ahead}}^* = 0.4$ as dashed reference lines. (b) Q2: conditional ATE on the high-king-shield subset. (c) Q3: CATE by space-control tercile (false-positive check). Bars: median over 10 seeds; error bars: IQR.

where d is distance-to-goal, ρ is local obstacle density, δ is difficulty, s is start-distance, k is step number, and $\chi(S) = (r + c) \bmod 2$ is the checkerboard parity of the agent cell; subscript z marks a standardized copy. The outcome DGP is

$$Y(0) = 0.4\delta_z + 0.2s_z + 0.1k_z + 0.3d_z + 0.2\rho_z + 0.5\chi_z + \gamma U + \varepsilon_0, \quad Y(1) = Y(0) + \tau^*,$$

with $\tau^* = 0.3$, $U \sim \mathcal{N}(0, 1)$, $\varepsilon_0 \sim \mathcal{N}(0, 0.5^2)$, and a logit shift $\alpha_U = 0.3$ on T_{obs} . The parity χ is not in C , but it can be recovered from the one-hot S , so the checker term is a confounder that is always present and visible only through S . Here γ therefore sets the strength of *additional* latent confounding on top of the checker baseline.

The same regime structure appears. Concept wins at the smallest $N = 500$ at every γ (Claim 1), and raw overtakes at larger N (Claim 2). The crossover comes earlier than in chess: raw wins by $N=1,000$ when $\gamma \geq 0.3$, and by $N=2,000$ even at $\gamma=0$, because the checker signal is stronger than the residual structure in chess. Learned Z never wins a cell. As in chess, the treatment-orthogonal projection that defines Z removes the checker-discriminative direction of S , which in this DGP is also the most outcome-predictive direction of the residual.

Table 6: Q1: CATE by material balance. Heterogeneous τ^* , $\gamma = 0.3$, $N = 5,000$, median over 10 seeds. Planted truths: $\tau_{\text{behind}}^* = \tau_{\text{equal}}^* = 0.1$, $\tau_{\text{ahead}}^* = 0.4$.

Stratum	Rep	$\hat{\tau}$	SE	bias	n_{sub}
Behind ($m < -0.5$)	Raw	+0.225	0.331	0.153	1,571
	Concept	+0.178	0.036	0.078	1,571
	Learned	+0.218	0.048	0.118	1,571
Equal ($ m \leq 0.5$)	Raw	+0.302	0.182	0.202	2,358
	Concept	+0.183	0.027	0.083	2,358
	Learned	+0.201	0.034	0.101	2,358
Ahead ($m > 0.5$)	Raw	+0.496	0.423	0.199	1,063
	Concept	+0.515	0.041	0.115	1,063
	Learned	+0.549	0.054	0.149	1,063

Table 7: Q2: conditional ATE restricted to positions with $c_{\text{king_shield}} \geq 75\text{th percentile}$. Conditional query, not $do(c_{\text{king_shield}} = \text{high})$.

Rep	$\hat{\tau}$	SE	bias	n_{sub}
Raw	+0.349	0.258	0.236	1,733
Concept	+0.243	0.037	0.093	1,733
Learned	+0.170	0.044	0.026	1,733

L Implementation and Reproducibility Details

Compute. All experiments run on a single laptop. The full chess primary grid (2,700 fits) completes in ≈ 2 h wall clock; the gridworld grid (900 fits) in 20–30 min; the concept-query, conplus, nuisance, and ablation grids each in 5–20 min. No GPU is used.

Software stack. Python 3.11, NumPy 1.26, pandas 2.1, scikit-learn 1.4, LightGBM 4.3, EconML 0.15, python-chess 1.999+, PyArrow 15.0, matplotlib 3.8, seaborn 0.13. Pinned versions are in `pilot/requirements.txt`; the conda environment is `chess-pilot`. All scripts are resume-safe: each runner writes incrementally and skips cells whose $(N, \gamma, \tau\text{-regime}, \text{rep}, \text{estimator}, \text{outcome model}, \text{seed})$ key is already present.

Nuisance-model hyperparameters. The propensity model is logistic regression with L2 regularisation ($C=0.1$ for adjustment sets with $d > 200$, $C=1.0$ otherwise), `max_iter=5000`, `lbfgs`, wrapped in a `StandardScaler` pipeline. The LightGBM outcome model uses `n_estimators=200`, `max_depth=5`, `learning_rate=0.05`, `verbose=-1`. The linear outcome model used in the Claim 3 robustness swap is `sklearn.linear_model.Ridge` with `alpha=1.0`. All three reuse the same 10 seeds, so differences are attributable to the representation, not to split variance.

Learned representation. Z is computed by (i) standardising $[S; X]$, (ii) fitting a regularised logistic propensity, (iii) projecting out the 1-d propensity direction, and (iv) PCA-compressing the residual to 32 components. Z is fit once per N -sample per seed and reused across DR-Learner cross-fits. We accept this shortcut to keep run time manageable on a laptop.

Conplus variants. Both the PCA and T-PLS conplus variants regress S on C with a linear model, then compress the residual to 16 dimensions. PCA uses `sklearn.decomposition.PCA(n_components=16)`; T-PLS uses `sklearn.cross_decomposition.PLSRegression(n_components=16, scale=False)` fit with the observed treatment as target. Both then adjust on $[C, R, X]$, a 34-dimensional adjustment set.

DR-Learner configuration. All DR-Learners are `econml.dr.LinearDRLearner` with `cv=5`, `min_propensity=0.02` clipping, and per-cell seeds. Feature rebinding uses W (nuisance-only) rather than X (heterogeneity), giving each fit the full adjustment set as nuisance features. The

Table 8: Q3: CATE by space-control tercile (false-positive check).

Stratum	Rep	$\hat{\tau}$	SE	bias	n_{sub}
Low	Raw	+0.421	0.313	0.325	1,793
	Concept	+0.232	0.034	0.090	1,793
	Learned	+0.182	0.046	0.041	1,793
Mid	Raw	+0.424	0.272	0.268	1,774
	Concept	+0.239	0.032	0.084	1,774
	Learned	+0.234	0.043	0.077	1,774
High	Raw	+0.332	0.355	0.293	1,423
	Concept	+0.284	0.036	0.080	1,423
	Learned	+0.311	0.051	0.110	1,423

Fig 6 — Game-Outcome Sanity Ablation (Oracle-Correlation Defense)

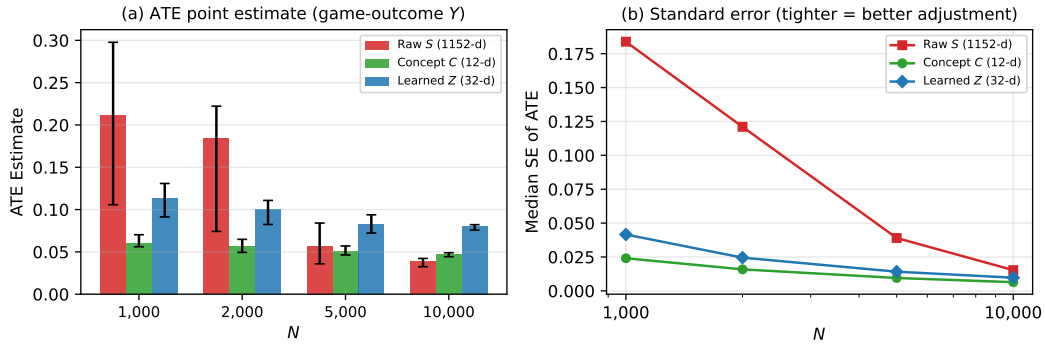


Figure 16: Game-outcome sanity ablation. (a) ATE estimate per representation across N (median with IQR). (b) Median standard error of the ATE per representation on a log-scale axis.

concept-query runner is the exception: it stratifies the subsample by the query variable and fits a separate DR-Learner ATE per stratum.

Seeds and reproducibility. Each runner uses NumPy `default_rng(1000+seed)` for sub-sample selection and `seed` directly for nuisance models. The 10 seeds are $\{0, 1, \dots, 9\}$. The gridworld pool is generated with seed 0; pool sub-samples use `default_rng(2000+seed)` to keep them independent of the chess sub-samples at the same seed index.

Grid totals. Chess primary: $5 \times 3 \times 2 \times 3 \times 3 \times 10 = 2,700$ fits. Gridworld: $5 \times 3 \times 1 \times 3 \times 2 \times 10 = 900$. Concept queries: $10 \times 3 \times 7 = 210$. Conplus (PCA and T-PLS): $5 \times 2 \times 10 = 100$ each. Nuisance diagnostics: $5 \times 1 \times 3 \times 10 = 150$. Game-outcome ablation: $4 \times 3 \times 2 \times 10 = 240$.

Committed artifacts. All decision caches, result CSVs, and figure PDFs are committed in `pilot/cache/`, `pilot/gridworld/cache/`, `pilot/outputs/`, and `../pilot/outputs/figures/`, so that every number, table, and figure in the main text and this appendix is reproducible without rerunning the experiments.

Table 9: Game-outcome ablation: DR-LightGBM ATE estimates and standard errors per representation, with relative efficiency ratio $SE(\text{raw})/SE(\text{concept})$. Medians over 10 seeds.

N	Raw S		Concept C		Learned Z		SE ratio raw/concept
	$\hat{\tau}$	SE	$\hat{\tau}$	SE	$\hat{\tau}$	SE	
1,000	+0.212	0.184	+0.061	0.024	+0.114	0.042	7.64
2,000	+0.184	0.121	+0.057	0.016	+0.100	0.025	7.62
5,000	+0.057	0.039	+0.051	0.010	+0.082	0.014	4.10
10,000	+0.039	0.015	+0.047	0.006	+0.081	0.010	2.39

Fig Gridworld — Secondary Environment Replication

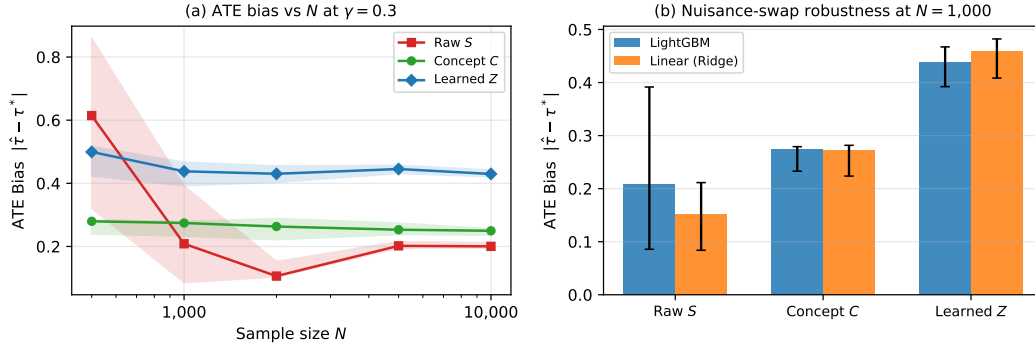


Figure 17: Gridworld replication. (a) ATE bias vs. N at $\gamma = 0.3$. (b) Nuisance-swap robustness bar chart at $N=1,000$, $\gamma = 0.3$. Concept wins at the smallest N , raw overtakes at larger N , learned Z underperforms throughout.

Table 10: Gridworld median ATE bias over 10 seeds, homogeneous τ^* , DR-Learner with LightGBM outcome. Bold: per-cell winner.

γ	N	Raw	Concept	Learned
0.0	500	0.540	0.183	0.402
0.0	1,000	0.292	0.204	0.362
0.0	2,000	0.047	0.185	0.368
0.0	5,000	0.136	0.183	0.383
0.0	10,000	0.136	0.176	0.365
0.3	500	0.615	0.280	0.499
0.3	1,000	0.209	0.274	0.438
0.3	2,000	0.106	0.263	0.430
0.3	5,000	0.202	0.253	0.445
0.3	10,000	0.200	0.250	0.430
0.6	500	0.580	0.354	0.518
0.6	1,000	0.276	0.326	0.497
0.6	2,000	0.190	0.344	0.509
0.6	5,000	0.273	0.318	0.508
0.6	10,000	0.271	0.321	0.497